



Faculty of Humanities and Social Sciences, University of Sri Jayewardenepura

Bachelor of Science Degree Second Year Second Semester Examination – Nov 2025

Information and Communication Technology

ICT 2233 Web Application Development

Duration: Check Below

Total Marks: 80

Q1. The final project consists of three (3) components, and you should submit the latter two separately.

- i. Topic Selection
- ii. System Report
- iii. Code Submission via GitHub Classroom

Topic Selection

All students are grouped (the list is available in the Learning Management System). Each group must follow the prerequisites below when selecting a topic:

- i. The system must have at least five main features. Count only top-level features. Sub-features do not increase the count. For example, “Authentication” is one main feature; its sub-features include “Login”, “Registration”, and “Password Reset”.
- ii. Provide at least five custom-developed endpoints implemented by the team. These must be student-written and non-trivial. The five endpoints must cover at least three distinct resources (for example, users, products, and orders).
- iii. The system must include a front end, a back end, and a database. You may use:
 - Next.js for both front end and back end, or
 - React for the front end with a simple Express.js based back end.

If you choose React with Express.js, you may keep both the frontend/ and backend/ folders in the same repository.

- iv. The system must be deployed and publicly accessible. You may choose any hosting provider. The database can be Vercel Postgres. If the application is hosted outside Vercel, connecting to Vercel Postgres remotely is acceptable.
- v. Continuous Integration and Continuous Deployment are optional and not assessed.

System Report Submission [20 Marks]

Prepare a report that explains the entire system clearly. Include all of the following:

- Cover Page
- Table of Contents
- Introduction
 - Motivation
 - Problem and Objectives
- System Overview, including capabilities and GUI
- Database Structure
- A detailed explanation of each API implementation and the respective endpoints
- Deployment description
- Summary

Formatting requirements (eligibility to be marked):

- Minimum number of pages: 15
- Body font size: 12 point
- Font: Times New Roman

Due Date of the Report: 20 November 2025 at 11:59 PM

System Report Marking Scheme [20 marks]

- **Introduction and Motivation** — 2 marks
- **Problem and Objectives** — 3 marks
- **System Overview and GUI** (diagram and core flows) — 3 marks
- **Database Structure** (schema and rationale) — 3 marks
- **APIs** (routes, methods, parameters, samples, errors, and access control) — 4 marks
- **Deployment description** (hosting, environment variables, database provisioning, and public URL) — 2 marks
- **Implementation Summary** — 2 marks
- **Conclusion and Writing Quality** — 1 mark

Code Submission [60 Marks]

Submit the codebase to GitHub Classroom. Exclude build artifacts and unnecessary folders or files in the final submission. Use a .gitignore file to exclude such items.

Accepted repository layouts:

- A single Next.js application at the repository root, or
- A repository with frontend/ and backend/ directories if using React with Express.js.

Repository root must include:

| -> (Next.js codebase, or frontend/ and backend/ folders if applicable)

| -> URL.txt (the single authoritative production URL)

| -> Postman API collection JSON (exported collection)

URL.txt must contain the publicly accessible URL of the hosted web application. This file is the authoritative source for the URL.

Additional requirements:

- The database must be hosted in Vercel Postgres.
- Marks are allocated strictly according to the rubric below.
- The hosted web application must be accessible from the internet using the URL in URL.txt.

Code Submission Marking Scheme (60 marks)

- **Features** (five main features, end-to-end with database) — 20 marks
- **UI and UX** (responsive, clear flows, usable) — 12 marks
- **APIs** (correct routes, status codes, and error handling) — 10 marks
- **Database** (sensible schema and constraints) — 6 marks
- **Deployment** (live, no runtime errors, and correct URL.txt) — 7 marks
- **Code Quality and Guidelines** (readability and standards) — 3 marks
- **Git Usage** (meaningful and regular commits) — 2 marks

Due Date: 5th of December 2025 at 11:59 PM

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